

Examiner Guide

■ Examiner's Guide – Final Exam

****Course:**** Applications of Large Language Models

****Exam:**** Programming Final (3 hours, 20% of grade)

****Base:**** `mrdbourke/simple-local-rag` (Colab version)

1. Purpose

Assess students' ability to ****apply**** prompting/decoding, ****modify**** retrieval, ****evaluate**** results, and ****reflect**** on AI/tool use. This is a capstone integrating Assignments 1–3.

2. Setup & Logistics

- Provide a pre-tested Colab notebook with dataset, baseline RAG, empty evaluation tables.
- Students submit: `.ipynb`, `.pdf`, and two CSVs.
- Time: 3 hours.

3. Sample Baseline Outputs (Typical)

- Q1: "Scotland, 1887." (Correctness 4, Coherence 4)
- Q2: "Bird impacts; minerals for solar." (3,3)
- Q3: "Energy storage helps." (3,2)
- Q4: Short, generic summary (3,3)
- Q5: "Hydropower due to displacement/ecosystems." (4,4)

4. Expected Student Work

Part A – Prompt & Decode

- Two prompt edits (persona/format/reasoning) + two decoding strategies.
- Evaluation table comparing baseline vs. modified across 5 queries.
- Clear commentary (2–3 sentences per change).

Part B – Retrieval

- ONE modification: chunking, embeddings, or simple re-rank.
- Evaluate precision/recall + relevance (1–4), complete table.
- Short commentary on trade-offs.

Reflections

- 3–4 sentence notes on tools used, reliability, and corrections made.

5. Grading Guide (Per Part = 50)

- ****Code**** 20: runs and implements required changes.
- ****Commentary**** 10: clear rationale.
- ****Evaluation**** 10: tables filled with notes.
- ****Reflection**** 10: thoughtful, concise tool log.

6. Common Pitfalls

- Overly terse notes; incomplete tables; omitted reflections; only cosmetic changes.

7. Benchmarks

- ****Excellent (90–100):**** Fully working, insightful commentary, strong evaluation, thoughtful reflections.
- ****Proficient (80–89):**** Minor gaps; generally complete and clear.
- ****Developing (70–79):**** Partial implementations; thin commentary; missing entries.
- ****Needs Improvement (<70):**** Non-functional; missing components; no reflection.

8. Bonus (+10) – Agentic Twist

- Self-Ask / Plan-and-Answer / Tool-Selector rule.
- Grading: Implementation 5, Commentary 3, Result 2.